

LinguistLLM Project Final Report

Introduction

Large Language Models (LLMs) have achieved state-of-the-art performance for machine translation of high-resource language pairs, but still substantially underperform for low-resource language pairs. This semester, I conducted joint research with Ammon Shurtz to see how prompt engineering and finetuning could potentially improve low-resource translation performance. The first experiment involved using prompt engineering with supplemental linguistic data to improve Finnish to English and Khmer to English translation. The second experiment involved finetuning LLAMA 2 on 5-shot Spanish to Quechua prompts and evaluating the finetuned model.

Experiment 1: Prompt Engineering with Linguistic Data

The goal of this experiment was to improve the translation quality of low-resource language pairs by providing useful linguistic data, such as dictionary entries and grammars, while prompting the LLM.

Methodology

For this experiment, we chose to use “Llama-2-7b-chat-hf” due to its multilinguality, being open source, and ease-of-use. We employed a retrieval-augmented generation (RAG) approach by searching for the relevant resources to insert into the prompt at inference time. We chose Finnish to English and Khmer to English as our experimental language pairs for a few reasons: (1) Ammon and I know enough Khmer and Finnish respectively to work with the data, (2) translation to English would be easier for the Student Research Conference (SRC) judges to interpret, and (3) we were able to source data for these pairs. Finnish to English is a medium-resource language pair since Finnish was included in LLAMA’s training data (roughly 0.03% of total). Khmer to English, on the other hand, is a low-resource language pair since

LLAMA has almost no Khmer in its training data. We ran 0-shot, 5-shot, dictionary-augmented, and grammar-augmented tests.

Data

Our translation segments were sourced from the FLORES-200 open dataset. Finnish and Khmer dictionaries were constructed using frequency lists from the Finnish IT Center for Science and Kheng.info web dictionary respectively. The terms from the frequency lists were then looked up in Wiktionary to create dictionaries of four thousand entries each. Grammars were initially sourced from Wikipedia, although they were too long for LLAMA’s context window. We ended up using shorter blog posts that described Finnish and Khmer grammar instead.

Results

We see no significant differences for Finnish to English translation between the different prompting methods. However, we do see major improvements for Khmer to English translation when integrating dictionary information, while grammar information seems to hurt translation ability. The large amount of grammar context appears to add too much noise to the input of the LLM. LLAMA 2 needs more linguistic information to be able to translate low-resource languages such as Khmer.

Experiment 2: Finetuning LLMs for Machine Translation

The goal of this experiment was to improve the translation quality of a low-resource language pair by finetuning an LLM for translating that specific pair.

Methodology

We decided to stick with “Llama-2-7b-chat-hf” due to its multilinguality and finetunability. We were able to utilize our prompt insertion script from the previous experiment

with slight modification to create our finetuning data. The LLM was finetuned on both 0-shot and 5-shot prompts. We intended to submit to the AmericasNLP shared task (although we did not in the end), so we chose Spanish to Quechua as our low-resource language pair since there were the most resources available of all the options. Spanish is also a language that Ammon and I can work with.

Data

AmericasNLP luckily provides a large dataset of parallel Spanish and Quechua data. After cleaning the data, we were left with 108,204 training pairs. All of these pairs were transformed into both 0-shot and 5-shot prompts using our modified prompt insertion script. For evaluation, we had 996 Spanish-Quechua pairs to compare our LLM output against.

Results

Unfortunately, complications arose in the finetuning of LLAMA 2 beyond one epoch and with multiple GPUs. Instead of finetuning LLAMA 2, we decided to run inference on our generated 5-shot prompts using the raw version of the newly released LLAMA 3. That way, we could get some preliminary results before the end of the semester. When prompted with our 5-shot Spanish to Quechua prompts, “Meta-Llama-3-8B-Instruct” produced translations with a chrF++ score of 15.99 over 996 test entries. The baseline for the same test set provided by AmericasNLP was 28.81 by Helsinki and 34.01 by Sheffield. While our results do not reach the level of the baseline, that is reasonable considering that LLAMA 3 probably has not been exposed to much if any Quechua before in its training.

Conclusion and Future Work

Our experiments show that large language models are capable of utilizing linguistic information to produce better translations. They are also suitable as translation systems for

low-resource languages, albeit with more effort. In the future, we hope to finish finetuning LLAMA 3 with our generated 0- and n-shot prompt data. We could also test finetuning LLAMA with our dictionary- and grammar-augmented prompts individually. If results are promising, we could then test combining the dictionary and grammar information together into a single prompt. With dictionary prompts, we could try to experiment with larger and smaller dictionaries to see how dictionary size correlates with quality improvements.

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